Cross-validation for the optimal choice of a spherical harmonic truncation order when reconstructing acoustic fields

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(Dated: 24 March 2026)

The sound field radiated or scattered by an object is often represented using spherical harmonics. The truncation order of the representation basis is usually chosen considering the equivalent size of the object, which is hard to define for most objects. Some authors proposed to optimally choose the truncation order that yields the lowest reconstruction error on a set of validation sensors, which demands extra measurement points. In this work, we propose to use cross-validation for the optimal choice of the truncation order. This allows to use each measurement point for both performing the interpolation and validating the interpolation accuracy. Cross-validation is applied to both simulated data and experimental data involving measurements on a controlled source with a cube-shaped array of 256 microphones. The proposed cross-validation metric is shown to be physically meaningful and to account for the order-limitation of the array used for the measurements.

[https://doi.org(DOI number)]

[XYZ] Pages: 1–12

I. INTRODUCTION

The family of spherical harmonic functions is commonly used in physics for representing functions on the unit sphere (Williams, 1999). In audio and acoustics, it first gained popularity for representing Head Related Transfer Functions (HRTF) (Aussal *et al.*, 2013; Evans *et al.*, 1998; Rafaely and Avni, 2010; Zotkin *et al.*, 2009). It is also a convenient basis for the representation of the Directivity Function (DF) of sound sources (Ackermann *et al.*, 2021; Ahrens and Bilbao, 2020, 2021; Pörschmann and Arend, 2021). The projection of the spatial dependency of a HRTF or DF not only allows to compress the spatial data, but also to store it independently of the measurement grid (Majdak *et al.*, 2022). Spherical harmonic representations of HRTFs and DFs are also commonly used in virtual acoustic simulations (Bilbao *et al.*, 2019; Helmholz *et al.*, 2019).

The spherical harmonic and spherical wave decomposition (respectively SH and SWD) of a HRTF or DF is generally identified using surrounding measurements via a least-squares inversion of a linear problem (Rafaely, 2015). This procedure is referred to in the literature as the Helmholtz Equation Least Squares (HELS) method (Wang and Wu, 1997). The truncation of the spherical harmonic basis when using this method is a subtle problem when working with irregular spherical or even non-spherical grids of transducers, and the acoustic literature lacks automatical procedures for choosing an adequate spherical wave truncation order.

In general physics, a common approach is to truncate the expansion at a spherical order $N = kd$, k and s being respectively the acoustic wavenumber and the extent of the source. In acoustics, this was first applied when analyzing the radiation of violins (Weinreich, 1985). Recent studies on both HRTF (Zhang *et al.*, 2010) and DF (Bellows and Leishman, 2023) interpolation made use of this rule. The identification of an adequate extent for the source of interest remains a delicate problem. In particular, the results from Bellows and Leishman 2023 evidence the existence of two different radii when analyzing the directivity of human speakers, respectively associated to the diffraction of sound by the head and by the torso.

Some authors (*e.g.* Wu 2015 when reconstructing the field radiated by a loudspeaker, and Bau *et al.* 2022 when interpolating HRTFs from measurements on irregular spherical grids) have proposed to choose the truncation order optimally, by assessing the ability of the truncated reconstruction basis to fit the measured data. This is generally done by using part of the measurement data to interpolate the acoustic field, and the remaining part to evaluate the fitness of the reconstruction. This approach is only feasible when the number of sampling points is sufficient for performing meaningful interpolation and fit quality assessment, therefore demanding ei-

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ther to deploy a higher number of transducers, or to perform extra measurements.

Structure of the document In this work, the use of cross-validation is proposed in order to identify an optimal truncation order for the SHD/SWD of a sound source's DF. This method enables to use all the transducers of an array dedicated to HRTF or DF measurements to serve as both interpolation and validation points. In Sec. II, the theory of the SWD and the principle of the HELS method are recalled. In Sec. III, existing procedures for defining an adequate truncation order are first recalled, before presenting how cross-validation can be used to identify an optimal SWD truncation order. In Sec. IV, the cross-validation framework is tested in numerical simulations. Section V presents an experimental validation of the proposed framework. Finally, Sec. VI concludes the present paper.

Mathematical and physical notations spherical coordinate system (r, θ, ϕ), exponentiel e, spherical harmonic functions (resp. associated Legendre function) of order $n \in [0, \ldots, \infty]$ and degree $m \in [-m, m]$ Y_{nm} (resp. P_n^m), spherical Hankel function h_n, δ_{nm} is the Kronecker function, inverse $^{-1}$ and hermitian transposition $.^H$, ceiling operator $\lceil . \rceil$. Vectors (resp. matrices) are denoted as bold (resp. and capital) letters. $||.||_2$ is the ℓ_2-norm on a given vector space. ρ_0 and c_0 respectively denote the density and speed of sound in air.

II. DIRECTIVITY RECONSTRUCTION USING THE HELS METHOD

A. The Spherical Wave Decomposition (SWD)

An harmonic sound field of frequency f (or equivalently wavenumber $k = 2\pi f / c_0$, where c_0 is the speed of sound) around a source centered at the origin of a spherical coordinate system (r, θ, ϕ) can be represented by a sum of spherical waves

$$P(r, \theta, \phi, k) = \sum_n \sum_m c_{mn} Y_n^m(\theta, \phi) h_n(kr), \qquad (1)$$

where $Y_n^m(\theta, \phi)$ are spherical harmonic functions of order n and degree m, $h_n(kr)$ are spherical Hankel functions of order n, and c_{mn} are the coefficients of the decomposition. In this work, we consider the DF measurement case, where an array of microphones placed at positions (r_q, θ_q, ϕ_q) surrounds the source. In the case where measurements are performed on a perfectly spherical surface of given radius r_0, the measured DF can be expanded as an infinite sum of spherical harmonic functions (Rafaely, 2015)

$$D_{r_0}(\theta, \phi, k) = P(r_0, \theta, \phi, k) = \sum_n \sum_m b_{mn}(k, r_0) Y_n^m(\theta, \phi), \qquad (2)$$

$$b_{mn}(k, r_0) = c_{mn} h_n(kr_0). \qquad (3)$$

In this work, we focus on the more general case where a spherical wave decomposition (SWD) is to be identified. This presents two advantages. First of all, the spherical harmonic decomposition (SHD) of the DF can be computed at any radius from the source using the SWD. Most importantly, the SWD allows to model the radial dependence of the sound field, enabling the identification of the SWD from measurements performed on non-spherical surfaces.

B. Identification of the SWD coefficients

Once a truncation order N is chosen, Eq. (2) can be written for each sensor $q \in [0, \ldots, Q-1]$ of the array. The vector $\mathbf{P} \in \mathbb{C}^Q$ of measured pressures can then be written

$$\mathbf{P} = \mathbf{H}\mathbf{c} + \mathbf{n}, \qquad (4)$$

where the elements of the noise vector $\mathbf{n} \in \mathbb{C}^Q$ follow independent centered circular normal random laws of variance σ^2. The coefficient at line q and column l in $\mathbf{H} \in \mathbb{C}^{Q \times (N+1)^2}$ is the spherical wave $\psi_{l(n,m)}(k, r, \theta, \phi) = h_n(kr) Y_n^m(\theta, \phi)$ indexed by $l(n, m) = n^2 + n + m$ (Ambisonic Number Sequence, see Chapman *et al.* 2009) and sampled at position q

$$\mathbf{H}_{ql} = [\psi_{l(n,m)}(k, r_q, \theta_q, \phi_q)]_q \qquad (5)$$

The coefficients of the truncated SWD can be estimated performing a regularized least-squares inversion of Eq. (4)

$$\hat{\mathbf{c}} = \mathbf{H}_\lambda^\dagger \mathbf{P}, \qquad (6)$$

where $\mathbf{H}_\lambda^\dagger = (\mathbf{H}^H\mathbf{H} + \lambda \mathbf{I})^{-1}\mathbf{H}^H$ is the Moore-Penrose inverse of the sensing matrix $\mathbf{H}$ ($.^H$ and $.^{-1}$ being respectively the Hermitian transposition and the inverse operator on matrices), and $\lambda \in \mathbb{R}^+$ is the regularization strength.

C. The need for regularization

1. Near-field unstability and ill-posedness

In certain configurations, inverse problems such as Eq. (4) can be ill-posed. To cope with this issue, a regularization term is generally introduced in Eq. (6) (refs). In acoustic imaging, ill-posedness typically arises when the different functions of the chosen wave basis (in our case, spherical waves) present high magnitude discrepancies on the measurement surface of radius r_{array}. In fact, at frequencies where the measurement points can be considered as taken in the far-field (high values of kr_{array}, *i.e.* at frequencies where the source-microphone distance is higher than the wavelength, typically above 100 Hz for an array of radius r_{array} =1 m radius), the spherical Hankel functions have all reached their $1/r$ far-field magnitude asymptote, ensuring that the inverse problem in Eq. (4) is well-posed. At low values of kr_{array}, Hartenstein *et al.* 2023 introduces the far-field truncation criterion. This criterion consists in including in the spherical wave basis only spherical waves that have reached their far-field behavior on the measurement surface at the frequencies of interest, ensuring that the inverse problem is well-conditioned. Such truncation is therefore a sufficient regularization method in the near-field case.

2. Experimental uncertainties and model mismatch

The unregularized least-squares SWD estimate (Eq. (6) with $\lambda = 0$) is robust to the presence of circular normal noise modeled by $\mathbf{n}$ in Eq. (4). In presence of experimental uncertainties such as sensor gain variability or microphone positioning uncertainty, pressure measurements deviate from the model in Eq. (4). In Hartenstein 2025, numerical simulations showed that the uncertainty in microphone positioning is particularly concerning at very high frequency where the wavelength is comparable to the positioning error, and that regularization is an efficient means for leveraging this model mismatch.

D. Reconstruction of the Far-Field Directivity Function (FFDF)

In the far-field ($kr \gg 1$), the Hankel functions behave as $j^{(n+1)} e^{-jkr}/r$ (Williams, 1999). As a result, the far-field pressure is simply the product of a point source radial dependence to a Far-Field Directivity Function (FFDF)

$$P(r, \theta, \phi, k) \approx \frac{e^{-jkr}}{r} D_\infty(\theta, \phi, f) \tag{7}$$

$$D_{\infty(\theta,\phi,f)} = \sum_n \sum_m j^{(n+1)} c_{mn} Y_n^m(\theta, \phi). \tag{8}$$

The FFDF D_∞ characterizes the angular dependence of the sound field measured sufficiently far away from the source. It is a radius-independent function that is useful in different applications (exemples et ref). It can be computed using the truncated SWD estimated via the HELS method.

E. Error on SWD estimate

To assess the quality of the reconstruction, one can compute the relative ℓ_2-error on SWD estimate

$$\mathcal{L}_{\mathrm{SH}} = \frac{||\hat{\mathbf{c}} - \mathbf{c}_{\mathrm{ref}}||_{SW,2}}{||\mathbf{c}_{\mathrm{ref}}||_{SW,2}}, \tag{9}$$

where the truncated SWD $\hat{\mathbf{c}}$ estimated using Eq.(6) is zero-padded up to the order of the reference SWD $\mathbf{c}_{\mathrm{ref}}$, and where $||.||_{SW,2}$ is the ℓ_2 norm operator on the space of SWDs.

The coefficients of a FFDF's truncated spherical harmonics decomposition are easily deduced from Eq. (8) as $\left(c_{mn}(k) j^{(n+1)}\right)_{n\in[0,N], m\in[-m,\ldots,m]}$. Following Parseval's theorem, the ℓ_2-norm $||D_\infty||_{\mathcal{S},2}$ of a FFD function D_∞ on the unit sphere is equal to the ℓ_2-norm $||\mathbf{c}||_{SH,2}$ of its spherical harmonic decomposition. This allows to establish an equality relation between $||D_\infty||_{\mathcal{S},2}$ and $||\mathbf{c}||_{SW,2}$

$$||D_\infty||_{\mathcal{S},2} = \sqrt{\sum_{n=0}^{N_{\mathrm{ref}}} \sum_{m=-n}^{n} |j^{(n+1)} c_{mn}|^2} = ||\mathbf{c}||_{SW,2}. \tag{10}$$

The same equation holds for a linear combination of FFDFs. Thus, the metric defined in Eq. (9) is equivalent to a relative ℓ_2 error on the unit sphere between the estimated and reference FFDFs, respectively denoted as $\hat{D}_\infty$ and $D_{\infty,\mathrm{ref}}$

$$\mathcal{L}_{\mathrm{SH}} = \frac{||\hat{D}_\infty - D_{\infty,\mathrm{ref}}||_{\mathcal{S},2}}{||D_{\infty,\mathrm{ref}}||_{\mathcal{S},2}}. \tag{11}$$

Equation (11) shows that the relative SWD error proposed in Eq.(9) is also equal to the ratio between the total energy of the FFDF estimation error on the unit sphere to the total energy of the FFDF on the unit sphere.

F. Identification of optimal parameters

The present work proposes to identify the truncation order that minimizes a pressure reconstruction error computed using the cross-validation procedure that will be described in Sec. III C. In fact, cross-validation can be used to jointly identify optimal values of the truncation order N and of the regularization parameter λ. However, solving such joint identification problem would be computationally costful. In Sec. IV B 1, numerical simulations will show that, sufficiently close to the optimal value of the regularization strength λ (optimality is here meant in terms of the cross-validation error, and will be referred to as CV-optimality in the following), the optimal truncation order does not critically depend on the regularization parameter. This allows for the prior identification of a CV-optimal regularization strength for each truncation order.

Several parameter-choice methods have been proposed for the choice of the regularization parameter in Eq. (6) (Hansen, 1998), and a plethora of studies have compared their performance in acoustic applications (see *e.g.* Gomes and Hansen 2008; Pereira *et al.* 2015). The Generalized Cross-validation (GCV) presents the advantage of yielding a regularization parameter that minimizes the cross-validation error (Hansen, 1998), for a reduced computational cost compared to traditional cross-validation.

Based on these comments, the following framework is proposed for the identification of CV-optimal regularization strength λ and truncation order N

1. First, GCV is applied to the prior identification of a CV-optimal regularization strength for each order that will be tested in the second step. For more information on the GCV principles, the interested reader can refer to Hansen 1998.

2. Then, the cross-validation procedure described in Sec. III C is applied to the inverse problem in Eq. (6) for different truncation orders using the priorly identified regularization parameters.

3. Finally, an optimality criterion that will be defined in Eq. (17) is used to choose the CV-optimal truncation order.

In turn, the SWD can be estimated using Eq. (6) and the identified CV-optimal parameters.

III. TRUNCATION OF THE SPHERICAL WAVE MODEL

A. The kd-rule

The Fourier decomposition of a function presents a decaying behavior with the order of the Fourier basis functions. In particular, the coefficients of the spherical harmonic decomposition of a HRTF or DF (or equivalently the coefficients of the SWD of the sound field radiated by a source or diffracted by an object) decay with the spherical wave order. In an early study on the directivity of violins, Weinreich 1985 recalls that the SWD coefficients of the sound field radiated by a source of finite extent d decay rapidly for orders higher than kd. In a study on the spherical harmonic representation of HRTFs, Zhang *et al.* 2010 show that the SHD of the sound field diffracted by an object of radius d can be truncated at order $\lceil ekd/2 \rceil$. More recently, Bellows and Leishman 2023 have shown that a source extent d can be identified by fitting a frequency-dependent order $N = kd$ fronteer above which most of the SWD energy is located. Note that this procedure needs to define an adequate energy threshold and assumes that a sufficiently high-order SWD is priorly available for the source under study.

B. Optimal truncation order

Although the kd-rule can be derived based on theoretical considerations, its use for the truncation of a SWD or SHD demands the definition of a source extent d. The aforementioned studies do not inform the reader on adequate definitions of these parameters. In some studies on the reconstruction of HRTFs and of radiated sound fields, the authors (respectively Bau *et al.* 2022 and Wu 2015) propose to define the truncation order optimally. In practical cases, the reference SWD $\mathbf{c}_{\text{ref}}$ is unknown, which prohibits the use of the error on SWD estimate $\mathcal{L}_{\text{SH}}$ as an optimality criterion. Instead, the authors of the aforementioned studies propose to choose the optimal order as the one that minimizes the least-squares reconstruction error on some additional sensors. Unlike the kd-rule, this data-based approach does not imply any extra parameter. However, this truncation strategy requires to sample the pressure field at additional measurement points, which can be cumbersome and costful in terms of installation and measurement time. In the remaining of this section, a framework is proposed in order to make use of each transducer of a surrounding array as both interpolation and validation data.

C. Cross-validation for the choice of an optimal truncation order

1. Random partitioning of the measurement set

Let $\mathcal{E} = [0, \dots, Q-1]$ be the set of Q pressure measurement points, and let's define S random subsets $\mathcal{E}_s$ of $\mathcal{E}$ such that the family $(\mathcal{E}_s)_{s\in[0,\dots,S-1]}$ forms a partition of $\mathcal{E}$, e.g.

$$\begin{aligned} &\forall\ s \in [0, \dots, S-1]\ \mathcal{E}_s \subset \mathcal{E} \\ &\forall\ s, s' \in [0, \dots, S-1],\ s \neq s' \Rightarrow \mathcal{E}_s \cap \mathcal{E}_{s'} = \emptyset. \\ &\cup_{s\in[0,\dots,S-1]}\ \mathcal{E}_s = \mathcal{E} \end{aligned} \tag{12}$$

The sets $\mathcal{E}_s$ will be referred to as validation subsets. In the remainder of this document, the $S = Q$ validation subsets are singletons containing one of the sensors used for the measurements. In this particular case, cross-validation is referred to in the literature as leave-one-out cross-validation.

2. Pressure reconstruction on a validation subset

For each validation subset $\mathcal{E}_s$, a vector $\mathbf{P}_s \in \mathbb{C}^{Q-1}$ of pressure measurements and a sensing matrix $\mathbf{H}_s \in \mathbb{C}^{(Q-1)\times(N+1)^2}$ for the HELS problem can be defined using the measurement points that do not belong to $\mathcal{E}_s$

$$\forall\ q \in \mathcal{E} - \mathcal{E}_s\ \ [\mathbf{P}_s]_q = p(r_q, \theta_q, \phi_q) \tag{13}$$

$$\begin{aligned} &\forall\ q \in \mathcal{E} - \mathcal{E}_s, \forall\ n \in \mathbb{N},\ \ \forall m \in [-n, \dots, n] \\ &[\mathbf{H}_s]_{q,l(n,m)} = [\psi_{l(n,m)}(k, r_q, \theta_q, \phi_q)]_q. \end{aligned} \tag{14}$$

This yields a vector $\hat{\mathbf{c}}_s$ of SWD coefficients that can be used to compute pressure values $\hat{p}(r_q, \theta_q, \phi_q)$ for each measurement point q of the validation subset $\mathcal{E}_s$.

3. Definition of the cross-validation error

Due to the properties of the partitioning, an unique value $\hat{p}(r_q, \theta_q, \phi_q)$ of reconstructed pressure is available at each measurement point (r_q, θ_q, ϕ_q) of the array. Eventually, a Root Mean Square (RMS) reconstruction error on the measurement points can be computed. In the following, the cross-validation error $\mathcal{L}_{\text{CV}}$ is defined as the RMS reconstruction error on the array normalized by the RMS pressure across the array

$$\mathcal{L}_{\text{CV}} = \frac{\sqrt{\sum_{q=0}^{Q-1} |p(r_q, \theta_q, \phi_q) - \hat{p}(r_q, \theta_q, \phi_q)|^2}}{\sqrt{\sum_{q=0}^{Q-1} |p(r_q, \theta_q, \phi_q)|^2}}. \tag{15}$$

The simulations and experiments presented in Secs. IV and V will show that, at a given frequency, a criterion can be defined to identify an optimal truncation order in terms of the cross-validation error $\mathcal{L}_{\text{CV}}$.

IV. NUMERICAL SIMULATIONS

To illustrate the properties of the cross-validation error, we hereby present a numerical example where the field radiated by a sound source of known SWD $\mathbf{c}^{\text{ref}}$ is measured by a surrounding spherical array of microphones.

A. Experimental conditions

1. Source model

The sound source of interest is a a uniformly vibrating cap of aperture 2 cm placed on a spherical baffle of radius 8.5 cm. This source model is commonly referred to in the literature as the spherical cap, and it admits the following analytical SWD (Williams, 1999) (U_{cap} is the cap's velocity, ρ_0 the density of air, and α the angular aperure of the cap)

$$\begin{aligned} c^{\mathrm{ref}}_{00}(k) &= -\alpha_0(ka)U_{\mathrm{cap}}(1-\cos\alpha) \\ c^{\mathrm{ref}}_{nm}(k) &= \alpha_n U_{\mathrm{cap}}(P^0_{n-1}(\cos\alpha) - P^0_{n+1}(\cos\alpha))\delta_{mn} \\ \alpha_n(ka) &= -j\frac{\rho_0 c_0}{h'_n(ka)}\frac{\sqrt{\pi}}{2n+1} \end{aligned} \tag{16}$$

2. Experimental setup

The experimental setup is depicted in Fig. 1, where the colored dots represent microphones and the gray sphere represents the source. The array consists of 256 microphones placed on a quasi-cube of approximate side 1.2 m. The source and microphone positions correspond to the ones of the experiment that will be presented in Sec. V, and were retrieved using an acoustic localization framework based on the RMDU algorithm (Vanwynsberghe *et al.*, 2019) and described in Hartenstein *et al.* 2025.

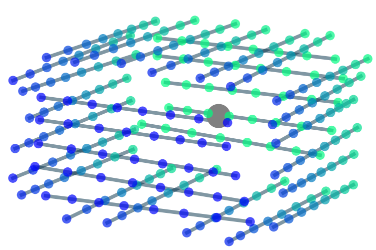


FIG. 1: Setup of the numerical simulations. Positions of the source (gray sphere) and microphones (color dots) used in the numerical simulations.

3. Synthetized sound field

Measurements are synthesized on the frequency range 100 Hz to 10 000 Hz using the analytical SWD from Eq. (16) and a high truncation order $N = 50$. At each frequency, the variance of the additive noise vector n is chosen 30 dB below the average Mean Squared pressure on the array (SNR = 30 dB). To account for microphone positioning uncertainty, each coordinate of the sensor positions used to compute the sensing matrix $\mathbf{H}$ are contaminated by a centered Gaussian noise of standard deviation $\Delta_x = 5/\sqrt{3}$mm. This results in a microphone positioning error $\Delta_r = \sqrt{3\Delta_x^2}$ of 5 mm, in accordance with the typical microphone positioning uncertainty measured in Hartenstein *et al.* 2025. Finally, in order to account for the gain variability of the sensors, the measured pressure value at each sensor position is multiplied by a factor $10^{\Delta_G/20}$, where Δ_G follows a centered Gaussian distribution of standard deviation σ_G =1 dB. This gain variability corresponds to the typical value measured in Hartenstein *et al.* 2024 for MEMS microphones of the same model as the ones used in the experiment presented in Sec. V.

4. Performance metrics

To account for the variability introduced by the different sources of uncertainty, the Monte-Carlo framework is used. 500 realizations of the scenario described in the last paragraph are run at a few frequencies (125, 250, 500 Hz, 1, 2,4, 6, 8, and10 kHz). The average (and standard deviation) of the relative SWD estimation error $\mathcal{L}_{\mathrm{SH}}$ are computed, and the optimal order $N^{\mathrm{opt}}_{\mathrm{SH}}$ is defined as the one that minimizes the average value of the SWD estimation error.

Running the cross-validation procedure for each of the realization would represent an untractable computational burden. In the following, the optimal order obtained with the cross-validation procedure will be shown to yield a SWD estimation error close to its optimal average value, leveraging the need for running cross-validation for each realization of the measured field.

B. Choice of the optimal parameters

1. Influence of regularization

Figures 2a, 2b, and 2c show the evolution with N of the cross-validation error with the truncation order N for several regularization strengths λ, at different frequencies. At low frequency (below 1 kHz), an analysis that is not presented in this paper showed that regularization has a negligible impact on the cross-validation error. At high frequency (4 kHz in Fig. 2a), regularization does not significantly improve the pressure reconstruction. This is in line with Sec. II C 1 : if the far-field criterion is satisfied and if the angular sampling is sufficiently dense in every direction, then regularization only has an impact at very high frequency, where the spherical model is significantly biased by the uncertainty in sensor positioning.

At very high frequency (6 and 8 kHz in Figures 2b and 2c), regularization allows for a significant decrease in cross-validation error. As recalled in Sec. II C 2, at very high frequency, the model mismatch due to the microphone positioning error being comparable to the wavelength can be mitigated by regularization. Moreover, the optimal truncation orders obtained with different regularization strengths close to the strength that yields a minimal cross-validation error are similar (see the orange, green, and red curves in Figs. 2b and 2c). This property ensures that the prior search of an optimal regularization strength proposed in Sec. II F yields

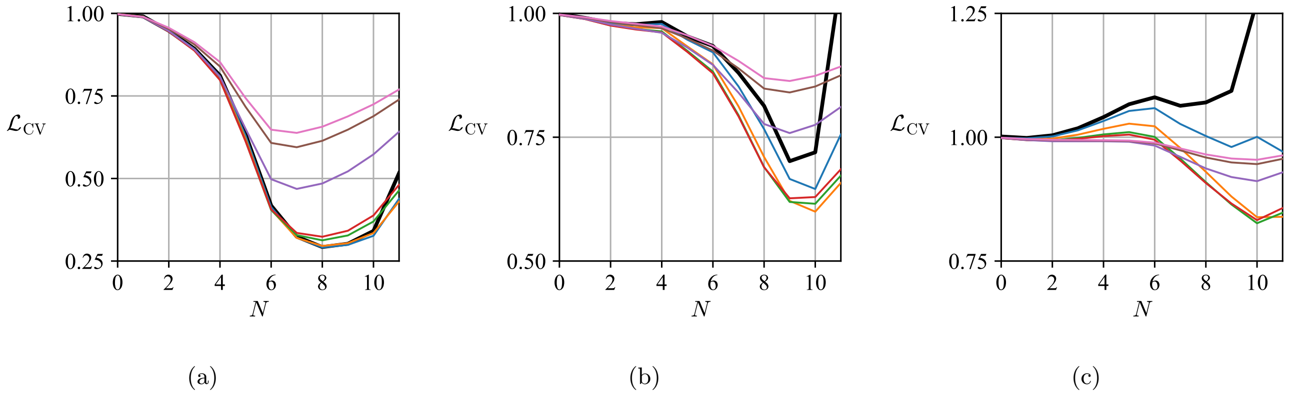


FIG. 2: Evolution of L_{CV} with N at (a) : 4 kHz(resp.(b), (c) : 6, and 8 kHz) for different regularization strengths λ indicated by different colors (black : $\lambda = 0$).

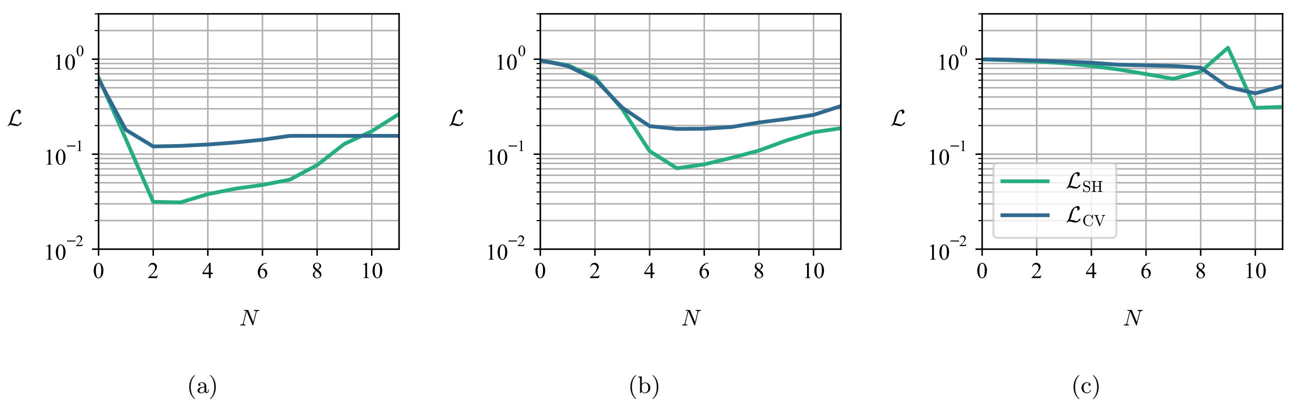


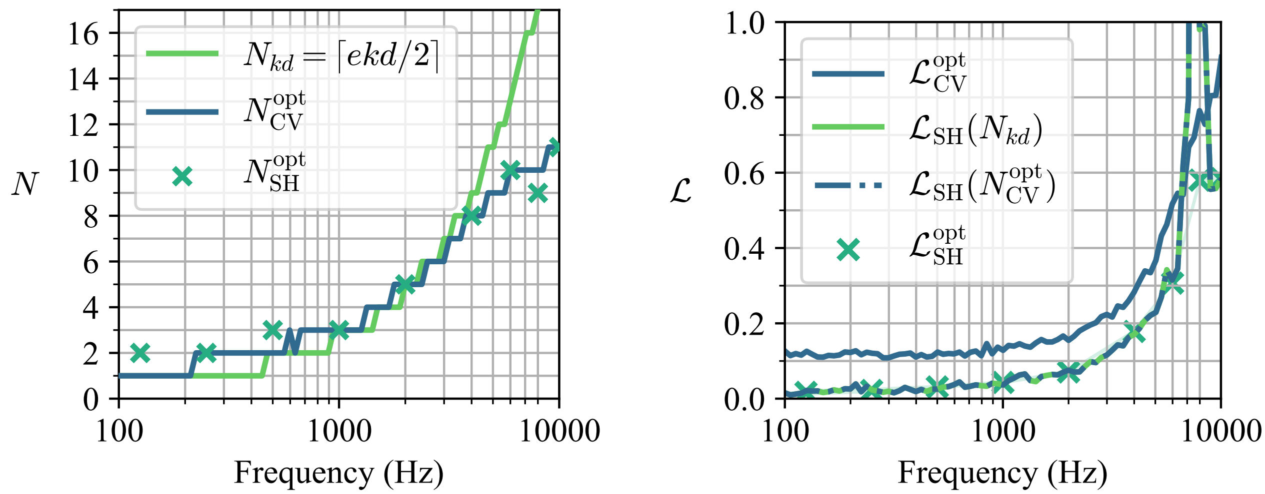
FIG. 3: Evolution of L_{CV} and L_{SH} with N at (a) (resp.(b), and (c)) 500 Hz (resp. 2, and 8 kHz) using the regularization parameter λ given by the GCV at each frequency and truncation order.

to similar results as if the optimal regularization strength and truncation order were sought for jointly.

2. Evolution of $\mathcal{L}_{\mathrm{CV}}$ and $\mathcal{L}_{\mathrm{SH}}$ with N

Figure 3 shows the evolution with truncation order N of the relative error on SWD estimate $\mathcal{L}_{\mathrm{SH}}$ and of the cross-validation error $\mathcal{L}_{\mathrm{CV}}$ at different frequencies. At all frequencies of interest and at low spherical wave orders, both error metrics decrease with the order. Once an order that is sufficient to represent the true spatial complexity of the sound field has been reached, the error metrics reach a lower plateau. The value of this plateau is higher for $\mathcal{L}_{\mathrm{CV}}$ than for $\mathcal{L}_{\mathrm{SH}}$. This is because the relative SWD estimation error only captures the error brought by the SWD estimation process, while the cross-validation error also includes the error brought by the pressure reconstruction using the estimated SWD. Neverthless, the minimal error stagnation orders are similar for both metrics, which highlights the ability of $\mathcal{L}_{\mathrm{CV}}$ to bring to light a minimal spherical wave order for capturing the spatial complexity of the sound field. At very high orders, $\mathcal{L}_{\mathrm{SH}}$ increases, which is a typical symptom of the overfitting phenomenon: high order spherical waves capture the unwanted spatial fluctuations of measured pressures brought by experimental uncertainties.

The high order increase of the cross-validation error appears at higher order and is less marked than the one of the SWD estimation error. This is because for dense samplings of the surface around the source, validation points are likely to be close to interpolation ones. In this case, if a slight overfitting occurs, the cross-validation error is likely to stay low for reasonable orders. It is only at very high orders (orders at which the minimal resolvable angle is lower than the angular distance between two microphones) that cross-validation allows to detect over-fitting. To prevent overfitting, the optimal cross-validation order $N_{\mathrm{CV}}^{\mathrm{opt}}$ is hereby defined as the lowest or-



(a) (b)

FIG. 4: Frequency evolution of (a) : N_{CV} and N_{SH} and (b) : $\mathcal{L}_{CV}$ and $\mathcal{L}_{SH}$.

der that yields close-to-minimal cross-validation error

$$N_{\mathrm{CV}}^{\mathrm{opt}} = \min_{N} N \quad \text{such that } \mathcal{L}_{\mathrm{CV}}(N) < \min_{N'} \mathcal{L}_{\mathrm{CV}}(N') + \epsilon. \tag{17}$$

The tolerance threshold ϵ should typically be fixed to a few percents (remind that $\mathcal{L}_{\mathrm{CV}}$ is defined relatively to the RMS pressure on the array), taking into consideration the fluctuations of $\mathcal{L}_{\mathrm{CV}}$ on its lower plateau. In the simulated and experimental studies presented in this document, ϵ is respectively fixed to 0.01 and 0.03.

C. Evolution of performance with frequency

1. Frequency evolution of the optimal orders

The frequency evolution of the optimal orders and of the order $N_{kd} = \lceil ekd/2 \rceil$ given by the kd-rule proposed by Zhang *et al.* 2010 is shown in Fig. 4a. Strikingly, the order $N_{\mathrm{SH}}^{\mathrm{opt}}$ that yields an optimal average value of the relative SWD estimation error $\mathcal{L}_{\mathrm{SH}}$ and the optimal cross-validation order $N_{\mathrm{SH}}^{\mathrm{opt}}$ defined in Eq. 17 are similar on the whole frequency range of interest. Expectedly, both optimal orders present an increasing trend with frequency. Up to a frequency of 4 kHz, the kd-rule yields truncation orders similar to the optimal ones. At very high frequencies, however, N_{kd} increases dramatically. This is because the kd-rule is only concerned by the radiation properties of the source, and not by the limitations of the array used for the measurement. On the other hand, the optimal cross-validation and SWD estimation criteria enable to quantify the loss of performance due to the spherical order needed to represent the radiated field exceeding the maximal order resolvable with the microphone array.

2. Frequency evolution of the errors

Fig. 4b represents the frequency evolution of the optimal cross-validation error $\mathcal{L}_{\mathrm{CV}}^{\mathrm{opt}}$ and SWD estimation error $\mathcal{L}_{\mathrm{SH}}^{\mathrm{opt}}$, and of the SWD estimation errors $\mathcal{L}_{\mathrm{SH}}\left(N_{\mathrm{CV}}^{\mathrm{opt}}\right)$ and $\mathcal{L}_{\mathrm{SH}}\left(N_{kd}\right)$ respectively obtained using the CV-optimal truncation order and with the truncation order N_{kd}. $\mathcal{L}_{\mathrm{SH}}(N_{\mathrm{CV}}^{\mathrm{opt}})$ and $\mathcal{L}_{\mathrm{CV}}^{\mathrm{opt}}$ present a monotonously increasing behavior with frequency. This behavior was observed in Hartenstein *et al.* 2023 and Hartenstein 2025, and is due to the uncertainty in microphone positioning becoming more and more comparable to the acoustic wavelength with increasing frequency. Moreover, as already mentioned and explained in Sec. IV B 2, $\mathcal{L}_{\mathrm{CV}}^{\mathrm{opt}}$ presents higher values than $\mathcal{L}_{\mathrm{SH}}^{\mathrm{opt}}$ at all frequencies.

Most importantly, the kd-rule and cross-validation criterion yield to relative SWD estimation errors $\mathcal{L}_{\mathrm{SH}}(N_{kd})$ and $\mathcal{L}_{\mathrm{SH}}(N_{\mathrm{CV}}^{\mathrm{opt}})$ almost identical as $\mathcal{L}_{\mathrm{SH}}^{\mathrm{opt}}$ up to 8 kHz. Around 9 kHz, $\mathcal{L}_{\mathrm{SH}}(N_{kd})$ and $\mathcal{L}_{\mathrm{SH}}(N_{\mathrm{CV}}^{\mathrm{opt}})$ present an accident **ok, à justifier plus tard ? j'attends les réponses de François et Fabrice**.

D. Reconstructed directivities

The two left columns in Fig. 5 represent the frequency-angle evolution of the theoretical directivity of the source under study (further left), and of the directivities reconstructed using the values of CV-optimal truncation order $N_{\mathrm{CV}}^{\mathrm{opt}}$ represented in Fig. 4a, in the horizontal (top) and vertical (bottom) planes. Up to 3 kHz, the reconstructed FFDF presents similar properties (om-

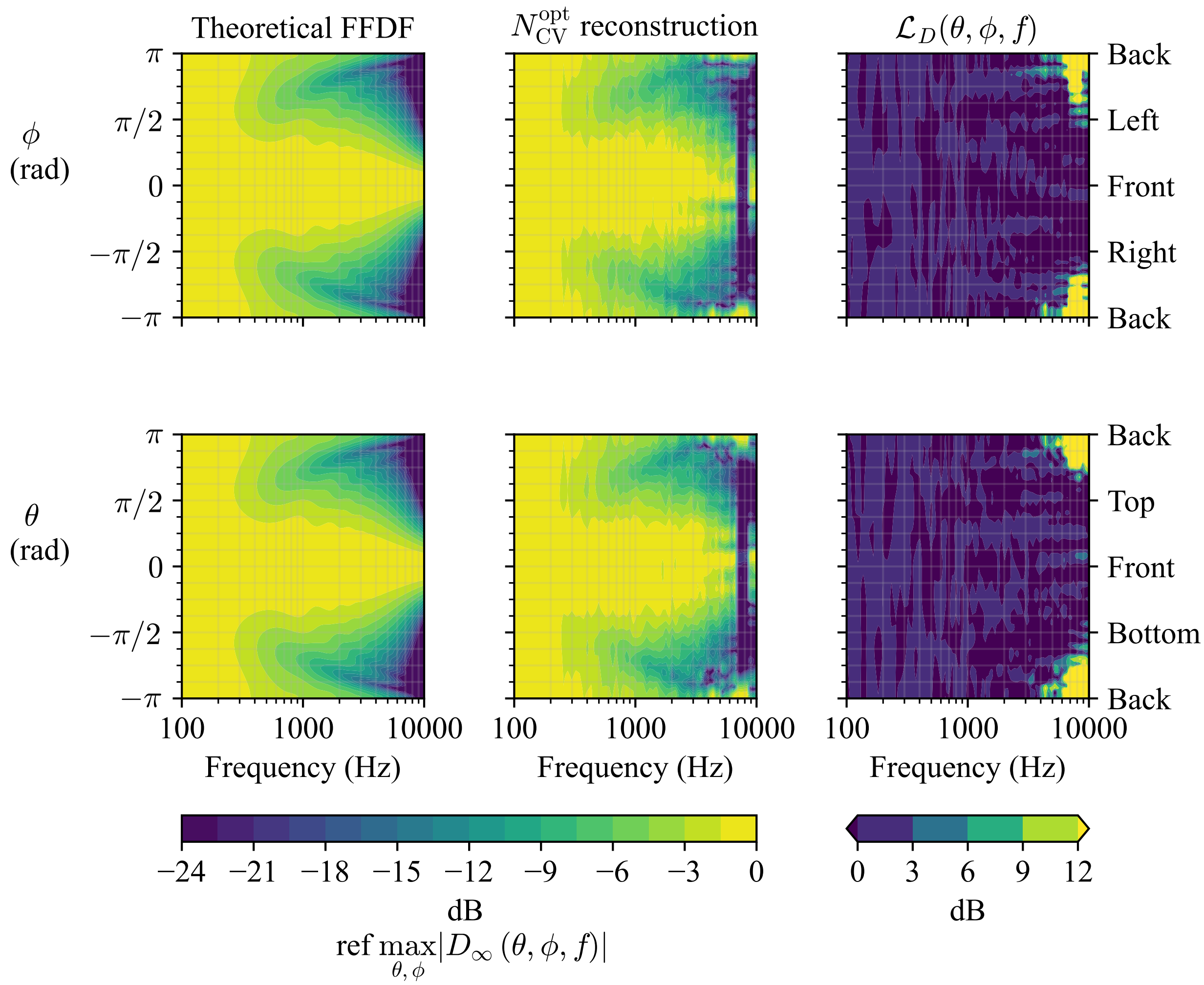


FIG. 5: Left and middle figures : Frequency-angle magnitude evolution of the theoretical spherical cap FFDF (left) and of the FFDF reconstructed using the CV-optimal truncation order (middle), shown in the horizontal (top) and vertical (bottom) planes. Right : Frequency angle evolution of the magnitude error $\mathcal{L}_D(\theta, \phi, f)$ defined in Eq. (18).

nidirectionality up to 200 Hz, main frontal lobe of varying width and rear and side lobes narrowing down with increasing frequency above 200 Hz) as the theoretical FFDF in both the horizontal and vertical planes (respectively top and bottom figures). From 3 kHz to 4 kHz, the reconstructed FFDF starts deviating from the theoretical spherical cap FFDF.

In order to further analyze the deviations between the theoretical and reconstructed FFDFs, the right column in Fig. 5 represents the frequency-angle evolution of the point-wise magnitude error between the reconstructed FFDF D_∞^{CV} and the theoretical FFDF D_∞^{ref}

$$\mathcal{L}_D(\theta, \phi, f) = 20 \log \left| \frac{D_\infty^{\mathrm{CV}}(\theta, \phi, f)}{D_\infty^{\mathrm{ref}}(\theta, \phi, f)} \right|. \tag{18}$$

$\mathcal{L}_D$ presents similar characteristics in the horizontal and vertical planes (respectively top and bottom figures). Up to approximately 3 kHz, the magnitude error is lower than 3 dB at all represented frequency and angles. At frequency higher than 3 kHz, horizontal side lobes of magnitude difference stronger than 6 dB appear at the rear of the source. The amplitude and number of these lobes increase with frequency. They are likely to result from spatial aliasing due to the high truncation orders involved (in Fig.4a, $N_{\mathrm{CV}}^{\mathrm{opt}} \geq 7$ to 8 at frequency higher than 3 kHz to 4 kHz) being not resolvable by the angular microphone sampling of the array.

V. EXPERIMENTAL VALIDATION

A. Setup

Experimental data were measured with a cube-shaped array of approximate side 1.2 m and 256 MEMS microphones installed in an anechoic chamber. The microphone signals were acquired using a homemade acquisition system developed at Institut Jean Le Rond ∂'Alembert and presented in Vanwynsberghe 2016. The sound source used in the experiments consists in a 3.5 cm radius speaker mounted on a 8.5 cm 3D-printed spherical baffle. The experimental setup is depicted in Fig. 6. The source and microphone positions shown in Fig. 1 and used to build the sensing matrix $\mathbf{H}$ were retrieved using the Robust Multi-Dimensional Unfolding (RMDU) algorithm from Vanwynsberghe *et al.* 2019 and the geometrical calibration framework proposed in Hartenstein *et al.* 2025.

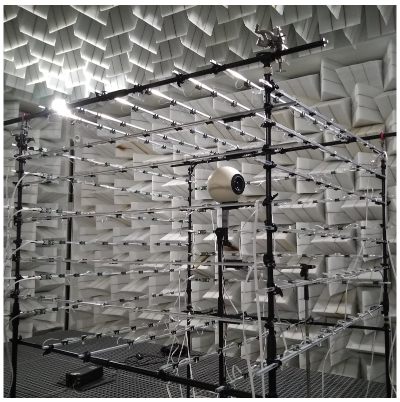


FIG. 6: Experimental setup.

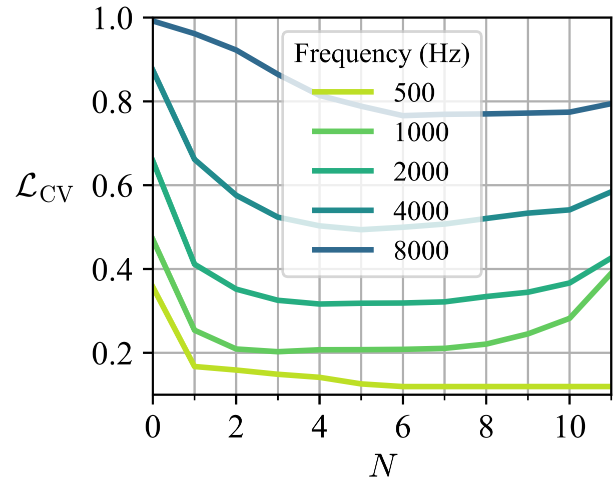


FIG. 7: Evolution of L_{CV} with N at different frequencies.

B. Identification of an optimal order

Figure 7 shows the evolution of the cross-validation error with truncation order N at different frequencies of interest. As in the simulated case analyzed in Sec. IV B 2, $\mathcal{L}_{\mathrm{CV}}$ starts decreasing at low orders to reach a lower plateau, and increases at higher orders. Note that, as in the simulated results shown in Fig. 3, close to minimal values of the cross-validation error are reached on a broad range of truncation orders N. This further justifies the use of the optimal cross-validation criterion proposed in Eq. (17) to define an adequate truncation order.

C. Performance analysis

The frequency evolution of the truncation orders $N_{\mathrm{CV}}^{\mathrm{opt}}$ and N_{kd} is represented in Fig. 8a. As in the simulated case shown in Fig. 4a, the optimal order $N_{\mathrm{CV}}^{\mathrm{opt}}$ identified with the cross-validation procedure presents an increasing trend with frequency, and becomes lower than N_{kd} from a certain frequency (approximately 1 kHz).

Fig. 8b shows the frequency evolution of the cross-validation error obtained with the optimal truncation orders shown in Fig. 8a. As in the simulated case (Fig. 4b), $\mathcal{L}_{\mathrm{CV}}$ presents an increasing trend with frequency, that is typically observed when reconstructing a sound field from noisy pressures sampled at uncertain positions. Moreover, the cross-validation errors obtained with experimental and simulated data present similar values.

D. Reconstructed directivities

Figure 9 represents the reconstructed FFDFs obtained with the truncation orders $N_{\mathrm{CV}}^{\mathrm{opt}}$ (left) and N_{kd} (center) shown in Fig. 8a. The reconstructed FFDFs present similar characteristics (poor directionality below 200 Hz with a slight left rear lobe, main frontal lobe from 500 Hz slightly tilted to the right, rear lobe of decreasing width above 500 Hz) up to 3 kHz. From 3 kHz, the FFDF reconstructed with the truncation order N_{kd} presents narrow lobes that do not appear on the CV-optimal reconstruction.

The right column of Fig. 9 shows the frequency-angle evolution (frequency range 1 kHz to 10 kHz) of the magnitude difference between the CV-*kd*-rule FFDF reconstruction D_{∞}^{kd} and the optimal FFDF reconstruction D_{∞}^{CV}

$$\Delta_D(\theta, \phi, f) = 20 \log \left| \frac{D_{\infty}^{kd}(\theta, \phi, f)}{D_{\infty}^{\mathrm{CV}}(\theta, \phi, f)} \right|. \qquad (19)$$

The magnitude difference between the reconstructions is lower than 3 dB at frequency lower than about 1.5 kHz. At frequency higher than 1.5 kHz to 2 kHz, it presents horizontal lobes of values higher than 6 dB. These lobes present increasing strength and number and decreasing width with increasing frequency from approximately 3 kHz. In this frequency range, the *kd*-rule yields truncation orders higher than 7, while the experimental CV optimal order is lower than 5 (see Fig. 8a).

In the simulated results presented in Sec. IV, Fig. 5, the reconstructed directivity was shown to present strong artifacts likely related to aliasing for truncation orders higher than 7 to 8. In a similar fashion, the magnitude difference in Fig. 9 reveals horizontal lobes that are

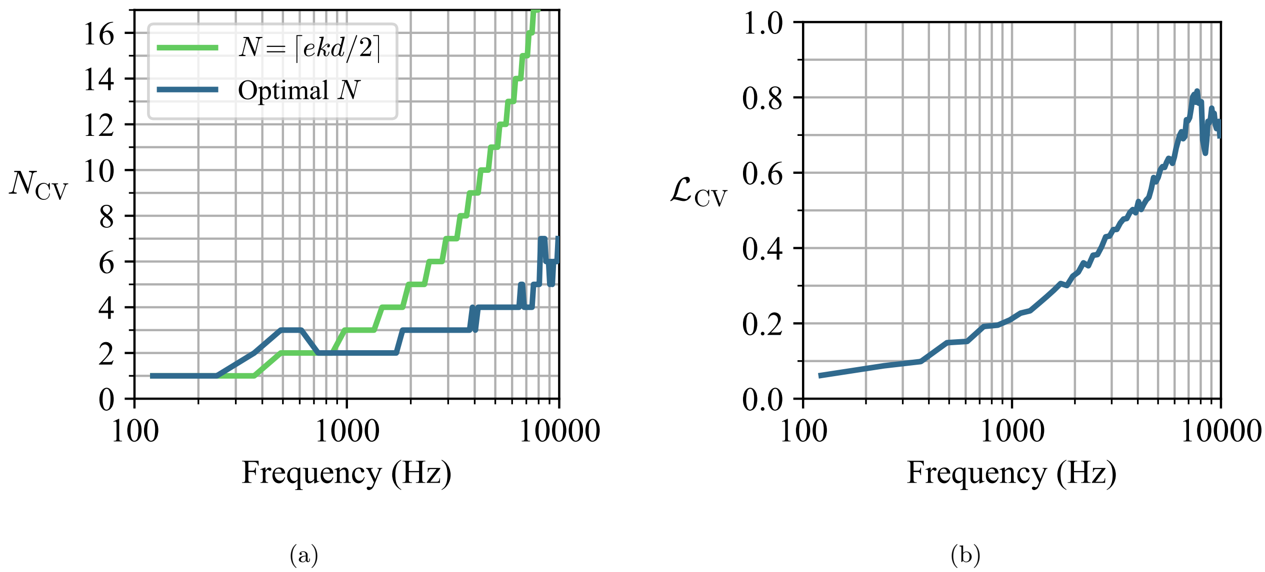


FIG. 8: (a) and (b) : Evolution of the optimal values of N_{CV} and L_{CV} with frequency.

stronger in the kd-rule FFDF reconstruction than in the CV-optimal FFDF reconstruction. As a conclusion, the cross-validation procedure proposed in this paper yields better robustness to aliasing effects than the kd-rule.

VI. CONCLUSION

This paper proposed to make use of cross-validation in order to identify the optimal truncation order associated with the SWD of the sound field radiated by an acoustic source.

Numerical simulations showed that the cross-validation metric defined in Eq. (15) enables to identify a close-to-optimal truncation order in terms of SWD estimation via the optimization problem in Eq. (17). While the commonly used kd-rule only takes into consideration the radiation properties of the sound source and yields excessively high truncation orders at high frequency, cross-validation highlights the order-limitation of the array used for the measurements. Additionally, the simulations showed that the regularization strength used in the SWD estimation process can be tuned priorly to the identification of an optimal truncation order, saving computational time in the cross-validation process.

The proposed cross-validation-based framework was applied to reconstruct the FFDF of a sound source based on measurements made with an array 256 MEMS microphones disposed on a cube-like surface. The cross-validation error was shown to enable the identification of an optimal truncation order. The obtained optimal order and cross-validation error were shown to present similar frequency evolutions as in the numerical simulations. Finally, the FFDF reconstructed using the kd-rule presented stronger aliasing artifacts than the one using the CV-optimal order. This last result further supports the idea that cross-validation intrinsically takes into account the order limitation of the array.

ACKNOWLEDGMENTS

This research was funded by the French National Research Agency (ANR-21-CE42-0017, 2021-25).

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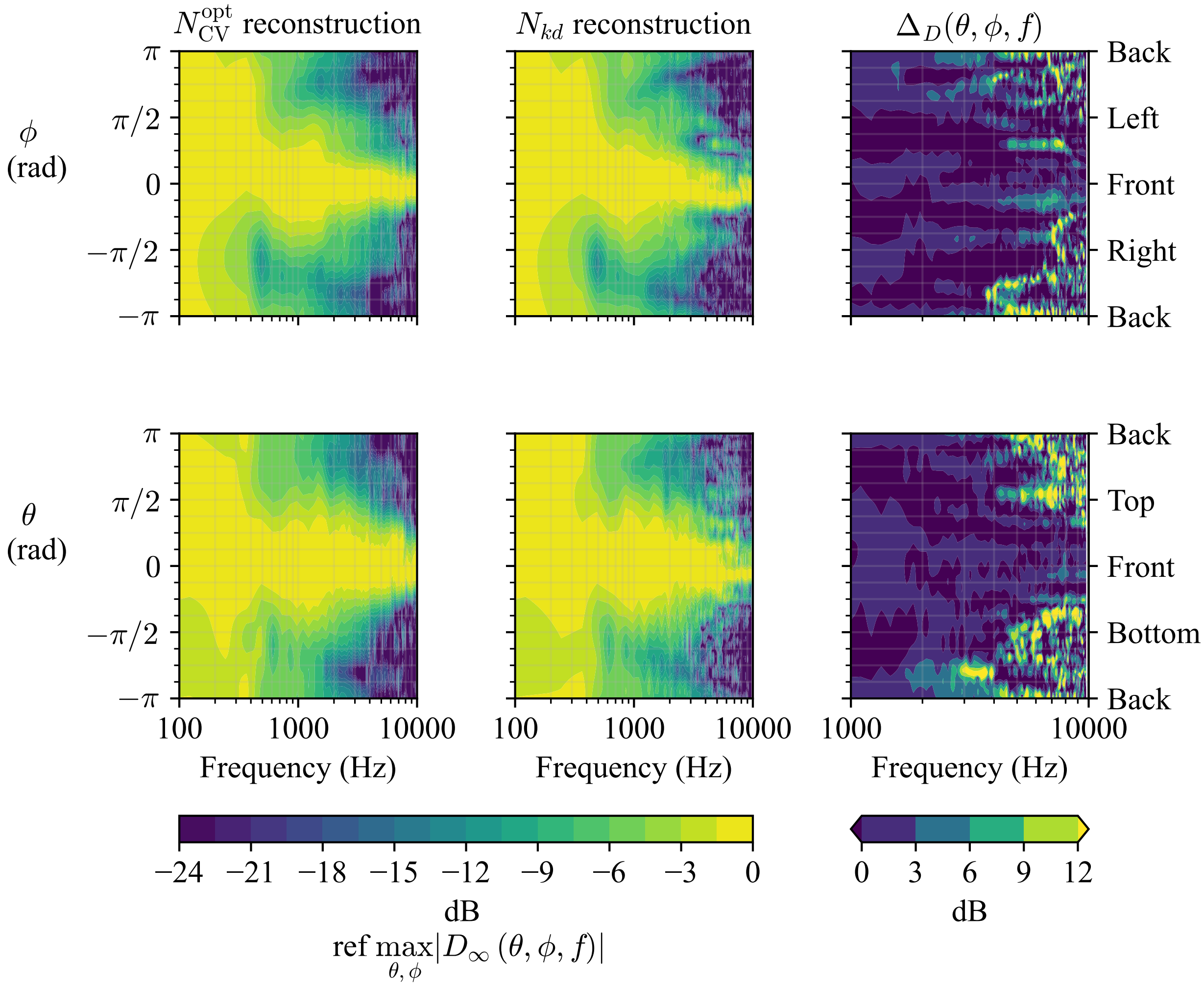

FIG. 9: Left and middle figures : Frequency-angle magnitude evolution of the FFDFs reconstructed using the CV-optimal truncation order (left) and the kd-rule (middle), shown in the horizontal (top) and vertical (bottom) planes. Right : Frequency-angle evolution of the amplitude difference $\Delta_D(\theta, \phi, f)$ defined in Eq. (19).

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